

Board of Studies (BoS) Syllabus Template

Course Information

Degree: UG

Program: BCA

Course Name: BCA

Course Credits: 3

Semester: 1

Course Category: Core

Type of Course: Lecture

New Course: Yes

Learning Hours

Activity	Hours per Week
Lecture	3
Tutorial	0
Practical	0
Experiential Learning	0

Course Contribution

Category	Percentage Contribution
Employment Enhancement	25%
Entrepreneurship Skills	30%
Skill Development	55%

Course Relevance

Level	Percentage Relevance
Global	30%
National	40%
Regional	30%
Local	30%

Course Outcomes (COs)

CO#	Course Outcome	Bloom's Taxonomy Level
CO1	Understand Ancient Education and learn the purpose and structure of Vedas, Vedangas, Upavedas, Puranas, and Itihasa.	BL-[2]
CO2	Explore Indian Knowledge Systems and identify and trace ancient wisdom.	BL-[3]
CO3	Apply Rational Solutions and use traditional knowledge for holistic well-being.	BL-[4]
CO4	Foster Pride & Sustainability and develop respect for Indian knowledge, human rights, and sustainable development.	BL-[5]

Module	Course Content	Tools Used / Assessment and Activity	Bloom's Level	Mapping COs	Percentage change in Module (put Nil if there is no change or it is a new course write NEW)
Module 1: Introduction to Indian Knowledge Systems (IKS)	Introduction to ancient Indian Knowledge Systems – Caturdaśa Vidyāsthānam, 64 Kalas, and Shilpa Śāstra. (1.5 hrs)		2	1	NEW
	Introduction to sacred Texts – Four Vedas, Vedāṅgas, and their significance. (1.5 hrs)		3	1	NEW
	Introduction to Indian Philosophical Systems (2 hours) Vedic (Sāṃkhya, Yoga, Nyaya, Vaiśeṣika, Pūrva-Mīmāṃsā, Vedānta) and Non-Vedic (Cārvāka, Buddhist, Jain)		4	1	NEW
	Ethics & Wisdom (1 hour)		3	1	NEW

	Niti Sastras and Subhasitas				
Module 2: Foundation Concepts for Science & Technology	1. Sanskrit Linguistics & Phonetics (1.5 hrs)– Study Panini’s Astadhyayi, verbs, and their computational relevance.		2	2	NEW
	2. Number Systems & Measurement (1.5 hrs) – Zero, place value, decimal system, and units for time, distance, weight		3	2	NEW
	3. Unique Number Representation (1.5 hrs) – Bhūta Saṃkhya, Kaṭapayādi systems, Pingala’s binary system		3	2	NEW
	Vaiśeṣika Philosophy (2 hours) Prameya, Pramāṇa, and Saṃśaya approach to physical reality		4	2	NEW
Module 3: Indian Mathematics & Astronomy in IKS	⑦ Mathematical Contributions –(2 hrs) Study the work of great mathematicians like those in Sulba Sutras, Aryabhatiya, and Pingala’s Chandah Sastra.		2	3	NEW
	⑦ Arithmetic & Geometry (1.5 hrs) – Explore arithmetic operations, geometry, trigonometry, and the value of π .		3	3	NEW
	⑦ Indian Astronomy (2hrs)– Learn about the celestial coordinate system and the Indian Calendar (Pancanga).		3	3	NEW

Module 4: Indian Science & Technology in IKS	Indian S&T Heritage & 64 Kalas (2 hrs)– Explore India’s rich science and technology heritage, including sixty-four art forms and occupational skills (64 Kalas).		2	4	NEW
	Metals & Metalworking Technology (2 hrs)– Study the use and technologies of Copper, Gold, Zinc, Mercury, Lead, and Silver in ancient India.		3	4	NEW
	Architecture & Town Planning (2 hrs)– Learn about town planning and architectural styles in India, with an emphasis on temple architecture.		3	4	NEW
Module 5: Humanities & Social Sciences in IKS	Health, Wellness & Psychology (2 hrs)– Explore Ayurveda, the role of sleep, food, and water in well-being, and the connection between body-mind-intellect-consciousness.		2	5	NEW
	Yoga & Well-being (1.5 hrs)– Understand the role of Yoga in promoting physical, mental, and spiritual wellness.		3	5	NEW
	Governance & Public Administration (2.5 hrs)– Examine the Indian approach to governance through texts like the Ramayana and Arthashastra.		3	5	NEW

CO-PO/PSO Mapping

PO #	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9
CO1	3	2	2	1	2	1	1	1	1
CO2	3	3	3	2	1	3	1	2	1

C O3	3	3	3	3	2	1	1	1	2
C O4	1	3	3	3	3	3	3	1	3
C O5	3	3	3	1	1	3	3	3	3

CO-PO Mapping for Indian Knowledge Systems (IKS)

CO #	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9
C01	✓	✓	✓	–	✓	–	–	–	–
C02	✓	✓	✓	✓	–	✓	–	✓	–
C03	✓	✓	✓	✓	✓	–	–	–	✓
C04	–	✓	✓	✓	✓	✓	✓	–	✓
C05	✓	✓	✓	–	–	✓	✓	✓	✓